The Backlot Blackout

Whodunnit? · Quadratic Functions & Complex Numbers · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Quadratic Functions & Complex Numbers	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Vertex form and completing the square, the quadratic formula and discriminant, complex numbers and complex roots, line and parabola systems, quadratic inequalities and projectile models.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Winston Hale took it, in the Prop Warehouse, using this method: Packed in a prop treasure chest.

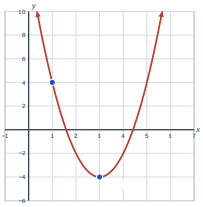
The reveal (read aloud): While the cast posed for photos, **Winston Hale** slipped into the **Prop Warehouse**, packed the Golden Reel inside a **prop treasure chest**, and had it carted off the lot with a truckload of "donated" props. Every alibi and method you cleared pointed straight at him. That's a wrap, detective!

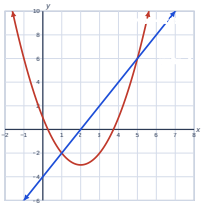
Clue map: what each correct clue clears

Card	Clears
1	Suspect: Marguerite Vance · Place: Editing Bay
2	Method: Hidden in a camera case
3	Suspect: Dashiell Cole · Place: Western Street Set
4	Method: Swapped for an empty film can
5	Suspect: Ines Okafor · Place: Costume Shop
6	Method: Rolled inside a costume trunk
7	Suspect: Bartholomew Pike · Place: Commissary
8	Method: Taped under a director's chair
9	Suspect: Lila Quintero · Place: Projection Booth
10	Method: Carried out in a popcorn tub

Answer Key

The Backlot Blackout · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	The graph shows a parabola. Its vertex and one other point are labeled. Write its equation in vertex form, $y = a(x - h)^2 + k$. 	**A.** $y = 2(x - 3)^2 - 4$	The vertex is $(3, -4)$, so $h = 3$ and $k = -4$. That gives $y = a(x - 3)^2 - 4$. Substitute the other point $(1, 4)$: $4 = a(1 - 3)^2 - 4$, so $4 = 4a - 4$. Then $8 = 4a$ and $a = 2$. $y = 2(x - 3)^2 - 4$	Marguerite Vance was reviewing the day's footage in the Editing Bay. The editor's keycard log shows her inside for the entire hour.
2	Complete the square to find the vertex of the parabola. Write the vertex as an ordered pair. $y = 2x^2 + 6x + 1$	**C.** $(-\frac{3}{2}, -\frac{7}{2})$	Factor 2 out of the x-terms: $y = 2(x^2 + 3x) + 1$. Half of 3 is $\frac{3}{2}$, and $(\frac{3}{2})^2 = \frac{9}{4}$. Adding $\frac{9}{4}$ inside the parentheses really adds $2 \cdot \frac{9}{4} = \frac{9}{2}$, so subtract $\frac{9}{2}$ outside: $y = 2(x^2 + 3x + \frac{9}{4}) + 1 - \frac{9}{2}$. $y = 2(x + \frac{3}{2})^2 - \frac{7}{2}$, so the vertex is $(-\frac{3}{2}, -\frac{7}{2})$	The camera case was opened at the security gate. It held only lenses, a tripod plate and a crumpled lunch napkin.
3	Solve with the quadratic formula. Give the exact solutions in simplest radical form. $3x^2 - 6x - 2 = 0$	**D.** $x = \frac{3 \pm \sqrt{15}}{3}$	Here $a = 3$, $b = -6$ and $c = -2$. The discriminant is $b^2 - 4ac = (-6)^2 - 4(3)(-2) = 36 + 24 = 60$. $x = \frac{-(-6) \pm \sqrt{60}}{2(3)} = \frac{6 \pm \sqrt{60}}{6}$. Simplify $\sqrt{60} = \sqrt{4} \cdot \sqrt{15} = 2\sqrt{15}$, so $x = \frac{6 \pm 2\sqrt{15}}{6}$. Divide every term in the numerator and the denominator by 2: $x = \frac{3 \pm \sqrt{15}}{3}$	Dashiell Cole was rehearsing a horse chase on the Western Street Set. Eight stunt riders and a wrangler saw him the whole time.
4	Use the discriminant to decide how many solutions the equation has and whether they are real or complex. $4x^2 - 12x + 9 = 0$	**C.** one real solution (a repeated root)	Here $a = 4$, $b = -12$ and $c = 9$. Square b first: $(-12)^2 = 144$, not -144. $b^2 - 4ac = 144 - 4(4)(9) = 144 - 144 = 0$. A discriminant of 0 means the two solutions are the same real number: one real solution (a repeated root)	The empty film can sat on its archive shelf, still sealed with the vault's numbered tape. Nobody swapped it.

5	Write the expression in **a + bi** form. $(3 - 2i)(4 + 5i) + i^{27}$	**B.** $22 + 6i$	Multiply with FOIL: $12 + 15i - 8i - 10i^2$. Since $i^2 = -1$, the last term is $+10$, so the product is $22 + 7i$. Powers of i repeat every 4: $i^{27} = i^{24+3} = (i^4)^6 \cdot i^3 = -i$. $(22 + 7i) + (-i) = 22 + 6i$	Ines Okafor was hemming a velvet cape in the Costume Shop. The seamstress next to her signed the work order at that minute.
6	Divide. Write the quotient in **a + bi** form with real and imaginary parts as fractions. $\dfrac{7 + i}{2 - 3i}$	**D.** $\dfrac{11}{13} + \dfrac{23}{13}i$	Multiply the top and bottom by the conjugate of the denominator, $2 + 3i$: $\dfrac{(7 + i)(2 + 3i)}{(2 - 3i)(2 + 3i)}$. Top: $14 + 21i + 2i + 3i^2 = 11 + 23i$. Bottom: $4 - 9i^2 = 4 + 9 = 13$. $\dfrac{11 + 23i}{13} = \dfrac{11}{13} + \dfrac{23}{13}i$	Every costume trunk was unpacked on the lawn. Only capes, top hats and a feather boa came out.
7	Solve over the complex numbers. Write both solutions in **a + bi** form. $2x^2 + 4x + 5 = 0$	**B.** $x = -1 \pm \dfrac{\sqrt{6}}{2}i$	$a = 2$, $b = 4$, $c = 5$. The discriminant is $4^2 - 4(2)(5) = 16 - 40 = -24$. $x = \dfrac{-4 \pm \sqrt{(-24)}}{4}$. Since $\sqrt{(-24)} = i\sqrt{24} = 2i\sqrt{6}$, $x = \dfrac{-4 \pm 2i\sqrt{6}}{4}$. Divide each term by 4 (the 4 and the 2 both reduce): $x = -1 \pm \dfrac{\sqrt{6}}{2}i$	Bartholomew Pike was eating chili at the Commissary. The cashier's receipt is stamped with the exact time of the blackout.
8	Find every point where the line meets the parabola. Write the points as ordered pairs. $y = x^2 - 4x + 1$ $y = 2x - 4$ 	**A.** (1, −2) and (5, 6)	Both equations equal y, so set them equal: $x^2 - 4x + 1 = 2x - 4$. Move everything to one side: $x^2 - 6x + 5 = 0$, so $(x - 1)(x - 5) = 0$ and $x = 1$ or $x = 5$. Find y with the line, $y = 2x - 4$: $y = -2$ when $x = 1$, and $y = 6$ when $x = 5$. The points are (1, −2) and (5, 6)	Each director's chair was flipped over and checked. Nothing was taped underneath any of them.
9	Solve the inequality. Write your answer in interval notation. $2x^2 - 5x - 3 \le 0$	**A.** $[-\dfrac{1}{2}, 3]$	Find the boundary points: $2x^2 - 5x - 3 = (2x + 1)(x - 3) = 0$, so $x = -\dfrac{1}{2}$ or $x = 3$. The parabola opens upward (the x^2 coefficient 2 is positive), so it is at or below the x-axis between the two roots. The inequality is $\le$, so both endpoints are included: $[-\dfrac{1}{2}, 3]$	Lila Quintero was running her lines in the Projection Booth. The projectionist's logbook has her signature across the hour.
10	A film crew fires a confetti canister straight up from a 64-foot soundstage roof. Its height in feet after t seconds is $h = -16t^2 + 48t + 64$. How many seconds after launch does the canister hit the ground? Ground level is $h = 0$	**D.** 4 seconds	Set the height to 0: $-16t^2 + 48t + 64 = 0$. Divide every term by -16: $t^2 - 3t - 4 = 0$. Factor: $(t - 4)(t + 1) = 0$, so $t = 4$ or $t = -1$. A time of -1 second is before the launch, so it does not fit the situation. The canister lands after 4 seconds	The giant premiere popcorn tubs were emptied onto a tarp. There was nothing in them but popcorn.

Wrong-Answer Map

The Backlot Blackout · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. $y = 2(x + 3)^2 - 4$	sign of h: the vertex x-coordinate is 3, so the factor is $(x - 3)$, not $(x + 3)$	Winston Hale was watching a rough cut in the Editing Bay. The film editor says he never left his seat.
1	C. $y = 2(x - 3)^2 + 4$	sign of k: the vertex y-coordinate is -4, so the constant is -4, not $+4$	Marguerite Vance was counting reels in the Prop Warehouse. The inventory clerk has her initials on the sheet at that minute.
1	D. $y = -4(x - 3)^2 - 4$	found a as rise over run, 8 over $-2 = -4$, forgetting to square $(1 - 3)$ before dividing	The prop treasure chest was pried open on the loading dock. It held only plastic doubloons and a rubber parrot.
2	A. $(\frac{3}{2}, -\frac{7}{2})$	sign of h: $x^2 + 3x + 9$ over 4 is $(x + 3$ over $2)^2$, so the vertex x-value is -3 over 2, not 3 over 2	Winston Hale was admiring a gown in the Costume Shop. The tailor chatted with him until the lights came back.
2	B. $(-3, 1)$	used $x = -b$ over a instead of $-b$ over $2a$ for the vertex x-value, then evaluated the function there	Security opened the treasure chest from the pirate set. It was lined with velvet and completely empty.
2	D. $(-\frac{3}{2}, -\frac{5}{4})$	subtracted 9 over 4 outside the parentheses, but adding 9 over 4 inside really adds $2 \cdot 9$ over $4 = 9$ over 2 because of the factor 2	Dashiell Cole was checking out sword props in the Prop Warehouse. The sign-out sheet has his name in green ink.
3	A. $x = 6 \pm \frac{\sqrt{15}}{3}$	divided only the square-root term by 6 and left the 6 in front undivided; the whole numerator must be divided	Ines Okafor was returning a hat box to the Prop Warehouse. The librarian stamped her return slip at that moment.
3	B. $x = \frac{6 \pm 2\sqrt{15}}{3}$	divided by $a = 3$ instead of $2a = 6$	Every prop treasure chest on the lot was tipped over and shaken. Only gold-painted rocks fell out.
3	C. $x = \frac{3 \pm \sqrt{3}}{3}$	forgot the double negative in $b^2 - 4ac$: with $c = -2$, $4ac = -24$, so the discriminant is $36 + 24 = 60$, not $36 - 24 = 12$	Winston Hale was sitting in the grandstand on the Western Street Set. Three extras say he never moved.
4	A. two real solutions	dropped the a in $4ac$: $144 - 4(9) = 108$, which is positive, so it looks like two real solutions; the correct $4ac$ is $4(4)(9) = 144$	Winston Hale was buying a sandwich at the Commissary. The cashier remembers him because he paid in coins.
4	B. no solutions	thought a discriminant of 0 means there are no solutions; it means the two solutions are the same real number	The treasure chest from the pirate set was weighed at the gate. It matched its listed weight exactly, so nothing was inside.
4	D. two complex solutions	evaluated $(-12)^2$ as -144, so the discriminant looked like $-144 - 144 = -288$; but $(-12)^2 = +144$	Bartholomew Pike was nailing crates together in the Prop Warehouse. A forklift driver waved to him through the window.
5	A. $2 + 6i$	treated i^2 as $+1$ in the multiplication, so the $-10i^2$ became -10 instead of $+10$	A crew member x-rayed the prop treasure chest at the clinic. The scan showed wood and nails and nothing else.
5	C. $22 + 8i$	wrote $i^{27} = i$ instead of $i^3 = -i$: 27 leaves a remainder of 3 when divided by 4, and $i^3 = -i$	Winston Hale was watching dailies in the Projection Booth. The projectionist wrote his name in the visitor log.

5	D. $21 + 7i$	used $i^{27} = -1$; but $i^2 = -1$ only for the exponent 2, and 27 leaves a remainder of 3 when divided by 4, so $i^{27} = i^3 = -i$	Lila Quintero was rehearsing a scene in the Prop Warehouse. The stage manager timed her through the whole blackout.
6	A. $\frac{11}{13} + 23i$	divided only the real part by 13; the imaginary part must be divided by 13 too	Marguerite Vance was picking out a candlestick in the Prop Warehouse. The prop master saw her there the entire hour.
6	B. $-\frac{11}{5} - \frac{23}{5}i$	wrote the denominator as $4 - 9 = -5$, forgetting that $i^2 = -1$ makes $(2 - 3i)(2 + 3i) = 4 + 9 = 13$	Winston Hale was giving notes to the editor in the Editing Bay. The editor typed every word into the project log.
6	C. $\frac{17}{13} - \frac{19}{13}i$	multiplied the top by $2 - 3i$, the denominator itself, instead of its conjugate $2 + 3i$, and kept 13 as the denominator	The big treasure chest from the pirate scene was unlocked and emptied. It contained only crumpled newspaper.
7	A. $x = 1 \pm \frac{\sqrt{6}}{2}i$	sign error on $-b$: with $b = 4$ the formula starts with -4, not $+4$	Police opened the treasure chest with a crowbar. Inside were sandbags and one very surprised mouse.
7	C. $x = -1 \pm \frac{\sqrt{14}}{2}$	added $4ac$ instead of subtracting it: $16 + 40 = 56$, so the solutions look real; the discriminant $16 - 40 = -24$ is negative	Dashiell Cole was loading a stunt wagon in the Prop Warehouse. The truck driver who helped him signed a receipt.
7	D. $x = -2 \pm i\sqrt{6}$	divided by 2 instead of $2a = 4$: $(-4 \pm 2i\sqrt{6})$ over 2 gives $-2 \pm i\sqrt{6}$ (the denominator is $2a$, which is 4)	Winston Hale was trying on a top hat in the Costume Shop. A seamstress took two photos of him at that very minute.
8	B. $(-1, -6)$ and $(-5, -14)$	sign error in the roots: $x^2 - 6x + 5 = (x - 1)(x - 5)$ gives $x = 1$ and 5, not -1 and -5	Winston Hale was shaking hands on the Western Street Set. The photographer has him in picture after picture.
8	C. $(5, 6)$	found only one intersection; the factored equation gives two values of x, and the graph shows the line crossing the parabola twice	Ines Okafor was hanging a costume on a rack in the Prop Warehouse. The clerk logged her entry just before the blackout.
8	D. $(-2, 1)$ and $(6, 5)$	wrote each ordered pair as (y, x); a point is (x, y), so $(1, -2)$ and $(5, 6)$	The prop treasure chest was dusted for prints and searched by hand. Not a thing was hidden inside.
9	B. $(-\frac{1}{2}, 3)$	used parentheses; the inequality is $\leq$, so the roots (where the expression equals 0) are included and need brackets	Bartholomew Pike was sanding a tabletop in the Prop Warehouse. Two apprentices worked beside him the entire time.
9	C. $(-\infty, -\frac{1}{2}] \cup [3, \infty)$	chose the outside region; the parabola opens upward, so it is at or below the x-axis between the roots, not outside them	Winston Hale was sharing a booth with a reporter in the Commissary. The reporter's recording has his voice the whole hour.
9	D. $[-3, \frac{1}{2}]$	sign error in the roots: $(2x + 1)(x - 3) = 0$ gives $x = -1$ over 2 and 3, not -3 and 1 over 2	A crew member lifted the lid on the prop treasure chest. It was packed with foam peanuts and nothing else.
10	A. -1 seconds	gave the negative solution; a time before the launch does not make sense, so it is rejected	Lila Quintero was signing autographs in the Prop Warehouse. A line of fans waited in front of her for an hour.
10	B. 1.5 seconds	found the time of the maximum height (the vertex), not the time the canister reaches the ground	Winston Hale was sipping tea in the Projection Booth. The projectionist poured it and noted the time.
10	C. 100 seconds	found the maximum height in feet (the y-value of the vertex) instead of the time in seconds	The treasure chest was carried to the sound stage and opened for the cameras. It was empty except for a fake map.